

Total No. of Questions : 4]

SEAT No. :

PF3

[Total No. of Pages : 2

APR-26/FE/Insem-3
F.E. (Common) (Insem)
ENGINEERING PHYSICS
(2019 Pattern) (Credit System) (Semester - II) (107002)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2 and Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicates full marks.
- 4) Assume suitable data if necessary.

Q1) a) Derive an expression for condition of maxima and minima for reflected light in case of uniform thickness thin transparent film. **[6]**

b) What is diffraction of light? Distinguish between Fraunhofer and Fresnel diffraction. **[5]**

c) How should the polarizer and analyser be oriented to reduce the beam of light to 25% of its original intensity? **[4]**

OR

Q2) a) Obtain the condition for principal maximum and minimum in fraunhofer diffraction due to a single slit. **[6]**

b) Explain double refraction phenomenon with the help of Huygen wave theory. **[5]**

c) A glass microscope lens ($\mu=1.5$) is coated with magnesium fluoride ($\mu_f=1.38$) film to increase the transmission of normally incident light of wavelength 5800AU. What minimum film thickness should be deposited on the lens? **[4]**

P.T.O.

- Q3)** a) Describe principal, construction and working of CO₂ Laser. [6]
- b) Define optical fibres. Explain step index fibre and graded index fibre. [5]
- c) Calculate the numerical aperture and acceptance angle of an optical fibre.
given : ($\eta_1 = 1.55$ and $\eta_2 = 1.50$) [4]

OR

- Q4)** a) Explain numerical aperture, acceptance angle and acceptance cone with required mathematical expression. [6]
- b) What is holography? Explain the process of hologram recording. [5]
- c) Explain : [4]
- i) Population inversion
 - ii) Resonant cavity

